

Cancer Staging & Classification

TNM System, AJCC Guidelines, and Pathological Grading

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1. Principles of Cancer Staging

Cancer staging provides a standardised language for describing disease extent, enabling prognostication, treatment planning, and communication across institutions. Staging systems are maintained by the American Joint Committee on Cancer (AJCC) and the Union for International Cancer Control (UICC), with editions updated approximately every 8 years based on accumulated outcome data. The 8th edition (2017) introduced significant refinements across 35 cancer sites.

Clinical staging (cTNM) is based on pre-treatment assessment; pathological staging (pTNM) incorporates surgical and histological findings and supersedes clinical staging where available.

2. TNM Classification System

The TNM system evaluates three anatomical parameters independently:

Parameter	Symbol Range	Description
Primary Tumour	T0–T4 (Tx)	Increasing local extent of primary tumour; Tis = carcinoma in situ
Regional Nodes	N0–N3 (Nx)	Absence or extent of regional lymph node involvement
Distant Metastasis	M0–M1 (Mx)	Absence or presence of distant metastases
Prefix c	cTNM	Clinical classification (pre-treatment)

Prefix p	pTNM	Pathological classification (post-surgery)
Prefix y	ycTNM / ypTNM	After neoadjuvant therapy
Prefix r	rTNM	Recurrent tumour after disease-free interval

3. Stage Groupings — Breast Cancer Example (AJCC 8th Ed.)

Stage	T	N	M	5-Year OS (approx.)
IA	T1	N0	M0	~99%
IIA	T0-2	N1 / N0 (T2)	M0	~93%
IIB	T2-3	N1 / N0 (T3)	M0	~80%
IIIA	T0-3	N2 / N3 (T3)	M0	~72%
IIIB	T4	N0-2	M0	~66%
IIIC	Any T	N3	M0	~54%
IV	Any T	Any N	M1	~28%

Note: Anatomic stage may be overridden by prognostic stage groups that incorporate ER/PR/HER2 status and tumour grade in breast cancer.

4. Pathological Grading

Histological grade reflects the degree of tumour differentiation and correlates with biological aggressiveness. Common grading schemes include:

- WHO 3-tier system (G1 well-differentiated, G2 moderately differentiated, G3 poorly differentiated) — used for most solid tumours
- Gleason Score (prostate) — primary + secondary pattern grades (2-10); now reported as Grade Groups 1-5
- Nottingham Grade (breast) — tubule formation + nuclear pleomorphism + mitotic count (score 3-9, grades I-III)
- ISUP grading (renal cell carcinoma) — based on nucleolar prominence
- FNCLCC system (soft tissue sarcoma) — differentiation + necrosis + mitotic rate

5. Molecular Classification

Molecular profiling increasingly supplements anatomic staging. Key examples include intrinsic breast cancer subtypes (Luminal A/B, HER2-enriched, Basal-like), CRC consensus molecular subtypes (CMS1-4), and WHO 2021 CNS tumour classification integrating IDH, 1p/19q, and TERT status. Multi-gene assays (Oncotype DX, MammaPrint) provide prognostic and predictive information beyond anatomic stage.

CLINICAL PEARL: In ER+ early-stage breast cancer, a low Oncotype DX Recurrence Score (<26 in postmenopausal women) supports omission of adjuvant chemotherapy per TAILORx trial data.